

VAPO: Value-Model-Based RL for Advanced Reasoning

60.4 on AIME 2024 — Surpassing DAPO by 10+ Points

ByteDance Seed | Base Model: Qwen2.5-32B | No SFT Data Required

Why Value Models? Finer Credit Assignment for Long CoT Reasoning

Value-Model-Free (GRPO, DAPO)

- Trajectory-level advantage computation
- Uniform credit assignment across all tokens
- Final reward propagates to entire sequence
 - Simpler but less precise

Value-Model-Based (VAPO)

- ✓ Token-level credit assignment
- ✓ Precise tracing of each action's impact
- ✓ Lower variance per token vs Monte Carlo
- ✓ Generalizes across samples for efficiency

Key Insight: Value models enable precise token-level credit assignment — critical when subtle step-level errors cause catastrophic failures in multi-step reasoning.

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

1 Value Model Bias

Bootstrapping from long trajectories compounds errors. Value estimates drift from true returns over extended reasoning chains.

2 Heterogeneous Sequence Lengths

Different response lengths require different bias-variance tradeoffs. One λ parameter cannot optimize both short and long sequences.

3 Reward Signal Sparsity

Only the final <eos> token receives reward. Long reasoning chains starve intermediate tokens of learning signals.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL



Challenges

VAPO Components →

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

Generalized Advantage Estimation (GAE) uses λ to balance bias vs variance: $\lambda=0 \rightarrow$ high bias, low variance (TD residual); $\lambda=1 \rightarrow$ low bias, high variance (Monte Carlo).

The Problem: Long-CoT reasoning produces responses from 100 to 10,000+ tokens. A single λ value cannot optimally trade bias for variance across this heterogeneity.

Short Responses (< 2K tokens)

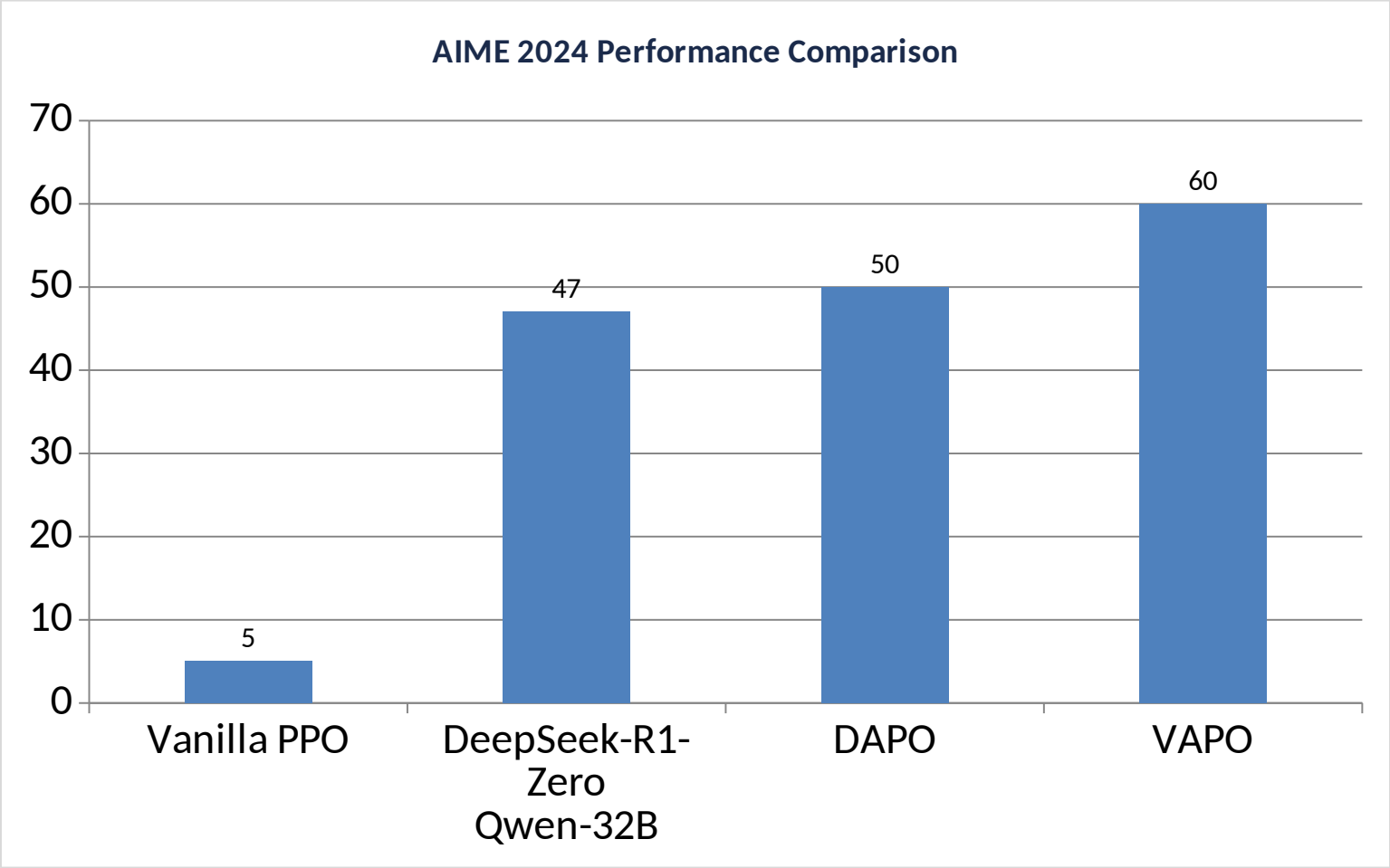
- Lower λ (0.1-0.3) preferred
- More TD variance reduction needed
- Bootstrap over fewer steps

Long Responses (> 5K tokens)

- Higher λ (0.7-0.9) preferred
- Bootstrap error accumulation risk
- Need low bias, can absorb variance

$$\lambda(\text{length}) = \text{clip}(0.1 + 0.8 \times (\text{length} - L_{\min}) / (L_{\max} - L_{\min}), 0.1, 0.9)$$

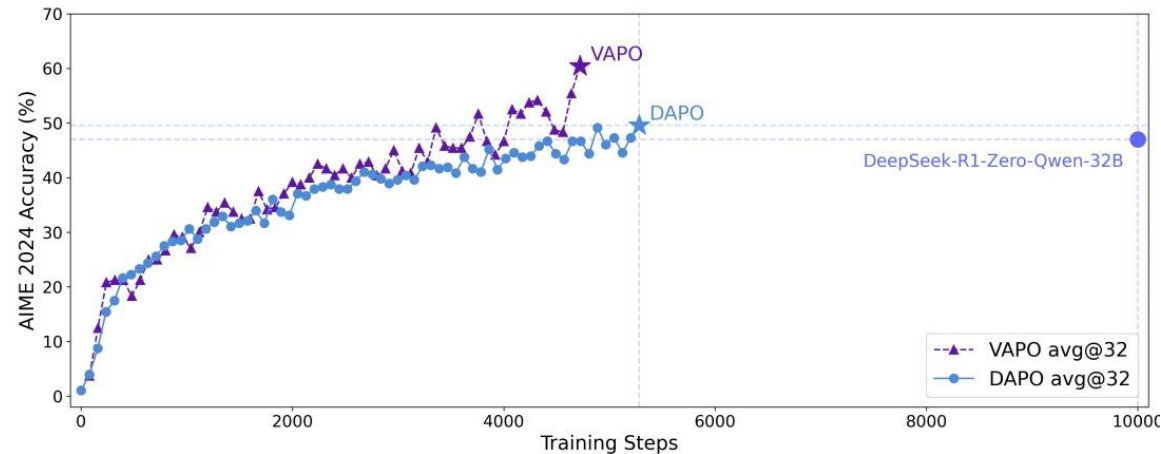
VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero



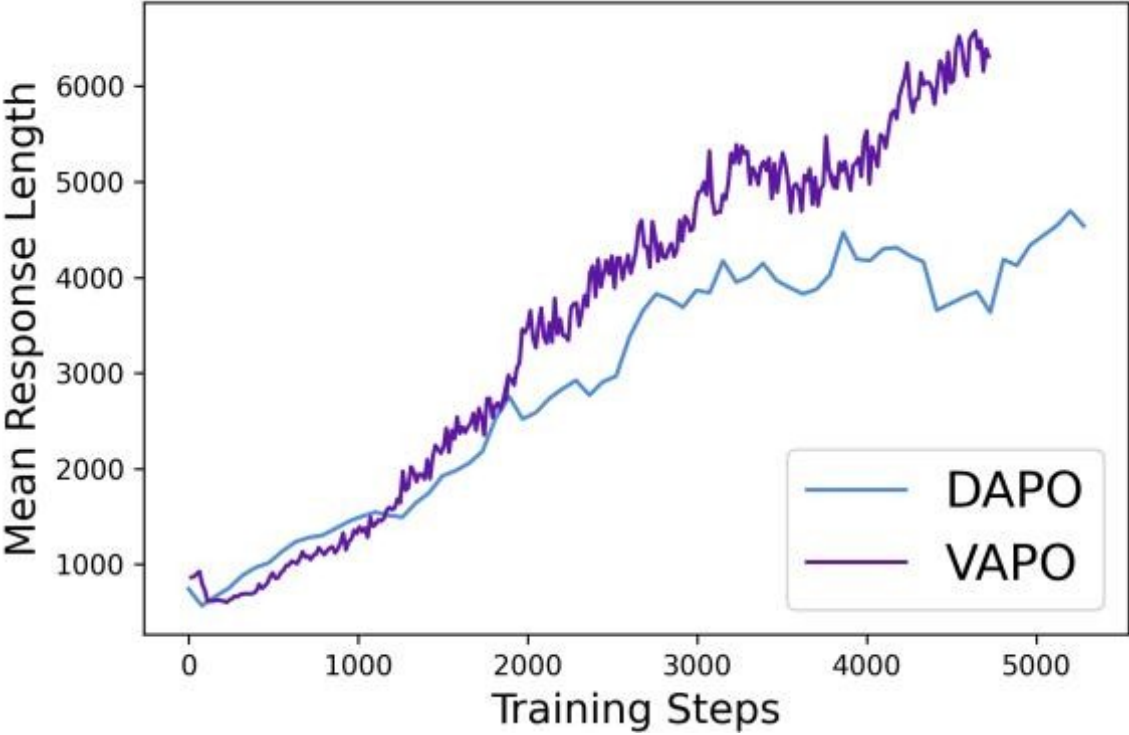
Key Results

- VAPO: 60.4 (avg@32)
 - DAPO: 50.0 (10.4 pts behind)
 - DeepSeek-R1-Zero: 47.0
 - Vanilla PPO: 5.0
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- Same base model: Qwen2.5-32B
 - No SFT data used
 - VAPO reaches SOTA in ~5K steps

VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO



AIME 2024 Accuracy vs Training Steps



Mean Response Length (tokens)

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Component Removed		
None (Full VAPO)		
Value-Pretraining		
Decoupled GAE		
Length-adaptive GAE		
Clip-Higher		
Token-level Loss		
Positive Example LM Loss		
Group-Sampling		

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

1

Value-model-based methods can surpass value-free methods when three core challenges (bias, heterogeneous lengths, reward sparsity) are systematically addressed.

2

Value-Pretraining is the single most impactful component — without it, value-model-based RL barely works (60 → 11).

3

Length-adaptive GAE is a novel and effective technique that addresses the heterogeneous sequence length problem unique to long-CoT reasoning.

4

VAPo is highly training-efficient and stable: reaches SOTA in ~5,000 steps with zero training crashes, vs DAPO's ~5,500 steps for a lower score.

5

Implications: Future reasoning model development should explore value-model-based RL with proper bias reduction and adaptive credit assignment.